RACE
Regular Analysis Through Continuous Exercise

Straight Line: RACE-01

Single Correct Type Questions:

1. A triangle with vertices (4, 0), (–1, –1), (3, 5) is
(A) isosceles and right angled
(B) isosceles but not right angled
(C) right angled but not isosceles
(D) neither right angled nor isosceles

MSL173

2. Find the value of x_1 if the distance between the points $(x_1, 2)$ and (3, 4) be 8.
(A) $3 \pm 2\sqrt{15}$ (B) $3 \pm \sqrt{15}$
(C) $2 \pm 3\sqrt{15}$ (D) $2 \pm \sqrt{15}$

MSL174

3. The four points whose co-ordinates are (2,1),(1,4),(4,5),(5,2) form :
(A) a rectangle which is not a square (B) a trapezium which is not a parallelogram
(C) a square (D) a rhombus which is not a square

MSL175

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4. If A and B are the points (–3,4) and (2,1), then the co-ordinates of the point C on AB produced such that $AC = 2BC$ are:
(A) (2,4) (B) (3,7)
(C) (7,–2) (D) $\left(-\frac{1}{2}, \frac{5}{2}\right)$

MSL176

5. If the line $2x + y = k$ passes through the point which divides the line segment joining the points (1, 1) and (2, 4) in the ratio 3 : 2, then k equals :
(A) $\frac{11}{5}$ (B) $\frac{29}{5}$
(C) 5 (D) 6

MSL177

Subjective Type Questions:

6. Find the distance between the points $P(-3, 2)$ and $Q(2, -1)$.

MSL178

7. If the distance between the points $P(-3, 5)$ and $Q(-x, -2)$ is $\sqrt{58}$, then find the value(s) of x.

MSL179

8. A line segment is of the length 15 units and one end is at the point (3, 2), if the abscissa of the other end is 15, then find possible ordinates.

MSL180

For More Join: @IITJEE_Advanced

9. Find the co-ordinates of the point dividing the join of $A(1, -2)$ and $B(4, 7)$:
(a) Internally in the ratio 1 : 2 (b) Externally in the ratio of 2 : 1

MSL181

10. In what ratio is the line joining $A(8, 9)$ and $B(-7, 4)$ divided by
(a) the point (2, 7) (b) the x-axis
(c) the y-axis.

MSL182

11. Determine the ratio in which the point $P(3, 5)$ divides the join of $A(1, 3)$ & $B(7, 9)$. Find the harmonic conjugate of P w.r.t. A & B.

MSL183

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ANSWER KEY

Question	1	2	3	4	5
Answer	A	A	C	C	D

6. $PQ = \sqrt{34}$ units **7.** $x = 6$ or $x = 0$ **8.** 11, –7
9. (a) (2,1) ; (b) (7,16) **10.** (a) 2 : 3 (internally); (b) 9 : 4(externally); (c) 8 : 7 (internally)
11. 1 : 2 ; Q (–5, –3)

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Straight Line: RACE-02

Single Correct Type Questions:

1. A triangle has a vertex at (1, 2) and the mid points of the two sides through it are (–1, 1) and (2,3). Then the centroid of this triangle is :

(A) $\left(\frac{1}{3},1\right)$ (B) $\left(\frac{1}{3},2\right)$

(C) $\left(1,\frac{7}{3}\right)$ (D) $\left(\frac{1}{3},\frac{5}{3}\right)$

MSL184

2. Coordinates of the vertices of a triangle ABC are (12,8), (–2,6) and (6,0) then the **correct** statement is -

(A) triangle is right but not isosceles

(B) triangle is isosceles but not right

(C) triangle is obtuse

(D) the product of the abscissa of the centroid, orthocenter and circumcenter is 160.

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MSL185

3. The orthocenter of the triangle ABC is 'B' and the circumcenter is 'S' (a,b). If A is the origin then the co-ordinates of C are :

(A) $(2a,2b)$ (B) $\left(\frac{a}{2},\frac{b}{2}\right)$

(C) $\left(\sqrt{a^2+b^2},0\right)$ (D) none

MSL186

4. The medians of a triangle meet at (0,–3) and its two vertices are at (–1,4) and (5,2). Then the third vertex is at -

(A) (4,15) (B) (–4,–15)

(C) (–4,15) (D) (4,–15)

MSL187

5. Let $A(-3,2)$ and $B(-2,1)$ be the vertices of a triangle ABC. If the centroid of this triangle lies on the line $3x+4y+2=0$, then the vertex C lies on the line

(A) $4x+3y+5=0$ (B) $3x+4y+5=0$

(C) $3x+4y+3=0$ (D) $4x+3y+3=0$

MSL188

For More Join: @IITJEE_Advanced

Subjective Type Questions:

6. The coordinates of the vertices of a triangle are (1, 2), (3, 4) and (4, 6) :

(a) Find centroid of the triangle.

(b) Find circumcentre & the circumradius.

MSL189

(c) Find orthocentre of the triangle.

MSL190

7. Line $\frac{x}{6}+\frac{y}{8}=1$ intersects the x and y axes at M and N respectively. If the coordinates of the point P lying inside the triangle OMN (where 'O' is origin) are (a, b) such that the areas of the triangle POM, PON and PMN are equal. Find

(a) the coordinates of the point P and

(b) the radius of the circle escribed opposite to the angle N.

MSL191

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5
Answer	B	D	A	B	C

6. (a) $\left(\frac{8}{3},4\right)$; (b) $\left(-\frac{7}{2},\frac{17}{2}\right)$, $\frac{5\sqrt{10}}{2}$, (c) (15, – 5) **7.** (a) $\left(2,\frac{8}{3}\right)$; (b) 4

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Straight Line: RACE-03

Single Correct Type Questions:

1. Let $A(h,k)$, $B(1,1)$ and $C(2,1)$ be the vertices of a right angled triangle with AC as its hypotenuse. If the area of the triangle is 1 square unit, then the set of values which 'k' can take is given by

(A) $\{-1, 3\}$ (B) $\{-3, -2\}$

(C) $\{1, 3\}$ (D) $\{0, 2\}$

MSL192

2. In a triangle ABC, coodinates of A are $(1, 2)$ and the equations of the medians through B and C are $x + y = 5$ and $x = 4$ respectively. Then area of ΔABC (in sq. units) is

(A) 5 (B) 9

(C) 12 (D) 4

MSL193

Subjective Type Questions:

3. Find the area of the triangle whose vertices are $A(1,1)$, $B(7, -3)$ and $C(12, 2)$

MSL194

4. Find the area of the quadrilateral whose vertices are $A(1,1)$ $B(7, -3)$, $C(12,2)$ and $D(7, 21)$

MSL195

5. Prove that the points $A(a, b + c)$, $B(b, c + a)$ and $C(c, a + b)$ are collinear
(By determinant method)

MSL196

6. The point A divides the join of $P(-5, 1)$ & $Q(3, 5)$ in the ratio $K : 1$. Find the two values of K for which the area of triangle ABC, where B is $(1, 5)$ & C is $(7, -2)$, is equal to 2 units in magnitude.

MSL197

7. The area of a triangle is 5. Two of its vertices are $(2, 1)$ & $(3, -2)$. The third vertex lies on $y = x + 3$. Find the third vertex.

MSL198

Ranker's Academy JEE

ANSWER KEY

1.	A	**2.**	B	**3.**	25 square units
4.	132 square units	**6.**	$K = 7$ or $\frac{31}{9}$	**7.**	$\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Straight Line: RACE-04

Single Correct Type Questions:

1. Let $O(0,0)$ and $A(0,1)$ be two fixed points. Then the locus of a point P such that the perimeter of ΔAOP is 4, is:

(A) $8x^2 - 9y^2 + 9y = 18$ (B) $9x^2 + 8y^2 - 8y = 16$

(C) $8x^2 + 9y^2 - 9y = 18$ (D) $9x^2 - 8y^2 + 8y = 16$

MSL199

2. If the equation of the locus of a point equidistant from the point (a_1, b_1) and (a_2, b_2) is $(a_1 - a_2)x + (b_1 - b_2)y + c = 0$, then the value of '$c$' is

(A) $\sqrt{a_1^2 + b_1^2 - a_2^2 - b_2^2}$ (B) $\frac{1}{2}\left(a_2^2 + b_2^2 - a_1^2 - b_1^2\right)$

(C) $a_1^2 - a_2^2 + b_1^2 - b_2^2$ (D) $\frac{1}{2}\left(a_1^2 + a_2^2 + b_1^2 + b_2^2\right)$

MSL200

3. Locus of centroid of the triangle whose vertices are $(a\cos t, a\sin t)$, $(b\sin t, -b\cos t)$ and $(1, 0)$, where t is a parameter, is

(A) $(3x+1)^2 + (3y)^2 = a^2 - b^2$ (B) $(3x-1)^2 + (3y)^2 = a^2 - b^2$

(C) $(3x-1)^2 + (3y)^2 = a^2 + b^2$ (D) $(3x+1)^2 + (3y)^2 = a^2 + b^2$

Parkers Academy JEE

MSL201

4. If $A(2,-3)$ and $B(-2,1)$ are two vertices of a triangle and third vertex moves on the line $2x + 3y = 9$, then the locus of the centroid of the triangle is :

(A) $2x - 3y = 1$ (B) $x - y = 1$

(C) $2x + 3y = 1$ (D) $2x + 3y = 3$

MSL202

5. Let $A(2,-3)$ and $B(-2,3)$ be vertices of a triangle ABC. If the centroid of this triangle moves on the line $2x + 3y = 1$, then the locus of the vertex C is the line

(A) $3x - 2y = 3$ (B) $2x - 3y = 7$

(C) $3x + 2y = 5$ (D) $2x + 3y = 3$

MSL203

Subjective Type Questions:

6. Find the locus of a variable point which is at a distance of 2 units from the y-axis.

MSL204

7. Find the locus of a point which is equidistant from both the axes.

MSL205

8. Find the equation to the locus of a point which moves so that its distance from the axis of x is three times its distance from the axis of y.

MSL206

For More Join: @IITJEE_Advanced

9. Find the equation to the locus of a point which moves so that the sum of the squares of its distances from the axes is equal to 3.

MSL207

10. Find the equation to the locus of a point which moves so that its distance from the point (3, 0) is three times its distance from (0, 2).

MSL208

11. Find the equation to the locus of a point which moves so that its distance from the axis of x is always one half its distance from the origin.

MSL209

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5
Answer	B	B	C	C	D

6. $x = \pm 2$ **7.** $y = \pm x$ **8.** $y = \pm 3x$ **9.** $x^2 + y^2 = 3$

10. $8x^2 + 8y^2 + 6x - 36y + 27 = 0$ **11.** $x^2 = 3y^2$

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Straight Line: RACE-05

Single Correct Type Questions:

1. If the perpendicular bisector of the line segment joining the points $P(1,4)$ and $Q(k,3)$ has y-intercept equal to –4, then a value of k is :-

(A) $\sqrt{15}$ (B) –2 (C) $\sqrt{14}$ (D) –4

MSL210

2. If in a parallelogram $ABDC$, the coordinates of A, B and C are respectively (1, 2), (3, 4) and (2, 5), then the equation of the diagonal AD is:-

(A) $5x + 3y - 11 = 0$ (B) $3x - 5y + 7 = 0$

(C) $3x + 5y - 13 = 0$ (D) $5x - 3y + 1 = 0$

MSL211

3. If the straight line, $2x - 3y + 17 = 0$ is perpendicular to the line passing through the points (7, 17) and (15, β), then β equals :-

(A) –5 (B) $-\frac{35}{3}$ (C) $\frac{35}{3}$ (D) 5

MSL212

4. If the two lines $x + (a-1)y = 1$ and $2x + a^2y = 1 (a \in R - \{0, 1\})$ are perpendicular, then the distance of their point of intersection from the origin is :-

(A) $\frac{2}{5}$ (B) $\frac{2}{\sqrt{5}}$ (C) $\frac{\sqrt{2}}{5}$ (D) $\sqrt{\frac{2}{5}}$

MSL213

5. A point on the straight line, $3x + 5y = 15$ which is equidistant from the coordinate axes will lie only in :

(A) 1st and 2nd quadrants (B) 4th quadrant

(C) 1st, 2nd and 4th quadrant (D) 1st quadrant

MSL214

6. If a line intercepted between the coordinate axes is trisected at a point $A(4,3)$, which is nearer to x-axis, then its equation is :-

(A) $3x + 8y = 36$ (B) $4x - 3y = 7$

(C) $x + 3y = 13$ (D) $3x + 2y = 18$

MSL215

7. If two vertices of a triangle are (5, –1) and (–2, 3) and its orthocentre is at (0, 0), then the third vertex is

(A) (4,–7) (B) (–4, –7) (C) (–4, 7) (D) (4, 7)

MSL216

8. If x_1, x_2, x_3 and y_1, y_2, y_3 are both in G.P. with the same common ratio, then the points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3)

(A) are vertices of a triangle (B) lie on a straight line

(C) lie of an ellipse (D) lie on a circle

MSL217

Rankers Academy JEE

For More Join: @IITJEE_Advanced

9. The diagonals of a parallelogram $PQRS$ are along the lines $x + 3y = 4$ and $6x - 2y = 7$. Then $PQRS$ must be a :

(A) rectangle (B) square

(C) cyclic quadrilateral (D) rhombus

MSL218

10. Number of lines that can be drawn through the point (4,–5) so that its distance from (–2,3) will be equal to 12 is equal to-

(A) 0 (B) 1 (C) 2 (D) 3

MSL219

Subjective Type Questions:

11. A line is such that its segment between the straight lines $5x - y - 4 = 0$ and $3x + 4y - 4 = 0$ is bisected at the point (1, 5). Obtain the equation.

MSL220

12. The points (1, 3) & (5, 1) are two opposite vertices of a rectangle. The other two vertices lie on the line $y = 2x + c$. Find c & the remaining vertices.

MSL221

13. Consider a triangle ABC with sides AB and AC having the equations $L_1 = 0$ and $L_2 = 0$. Let the centroid, orthocentre and circumcentre of the ΔABC are G, H and S respectively. $L = 0$ denotes the equation of sides BC.

(a) If $L_1: 2x - y = 0$ and $L_2: x + y = 3$ and $G(2, 3)$ then find the slope of the line $L = 0$.

MSL222

(b) If $L_1: 2x + y = 0$ and $L_2: x - y + 2 = 0$ and $H(2, 3)$ then find the y-intercept of $L = 0$.

MSL223

(c) If $L_1: x + y - 1 = 0$ and $L_2: 2x - y + 4 = 0$ and $S(2, 1)$ then find the x-intercept of the line $L = 0$.

MSL224

14. Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ & its third side passes through the point $(1, -10)$. Determine the equation of the third side.

MSL225

15. Two consecutive sides of a parallelogram are $4x + 5y = 0$ & $7x + 2y = 0$. If the equation to one diagonal is $11x + 7y = 9$, find the equation to the other diagonal.

MSL226

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	D	D	D	A	D	B	B	D	A

11. $83x - 35y + 92 = 0$

12. $c = -4; B(2, 0); D(4, 4)$

13. (a) 5; (b) 2; (c) 3/2

14. $x - 3y - 31 = 0$ or $3x + y + 7 = 0$

15. $x - y = 0$

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Straight Line: RACE-06

Single Correct Type Questions:

1. If a straight line passing through the point $P(-3, 4)$ is such that its intercepted portion between the coordinate axes is bisected at P, then its equation is :

(A) $x - y + 7 = 0$ (B) $3x - 4y + 25 = 0$

(C) $4x + 3y = 0$ (D) $4x - 3y + 24 = 0$

MSL227

2. The equation of the straight lines passing through the point $(4, 3)$ and making intercepts on the co-ordinates axes whose sum is -1 is

(A) $\frac{x}{2}-\frac{y}{3}=1$ and $\frac{x}{-2}+\frac{y}{1}=1$ (B) $\frac{x}{2}-\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

(C) $\frac{x}{2}+\frac{y}{3}=1$ and $\frac{x}{2}+\frac{y}{1}=1$ (D) $\frac{x}{2}+\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

MSL228

3. Area of the quadrilateral formed by the lines $|x| + |y| = 2$ is :

(A) 8 (B) 6 (C) 4 (D) none

MSL229

Pankers Academy JEE

4. If the x intercept of the line $y = mx + 2$ is greater than $1/2$ then the gradient of the line lies in the interval-

(A) $(-1,0)$ (B) $(-1/4,0)$

(C) $(-\infty,-4)$ (D) $(-4,0)$

MSL230

5. A line is drawn through the point $(1, 2)$ to meet the coordinate axes at P and Q such that it forms a triangle OPQ, where O is the origin. If the area of the triangle OPQ is least, then the slope of the line PQ is :

(A) $-\frac{1}{2}$ (B) $-\frac{1}{4}$ (C) -4 (D) -2

MSL231

6. If the x-intercept of some line L is double as that of the line, $3x + 4y = 12$ and the y-intercept of L is half as that of the same line, then the slope of L is :-

(A) -3 (B) $-3/2$ (C) $-3/8$ (D) $-3/16$

MSL232

7. A square of each side 2, lies above the x-axis and has one vertex at the origin. If one of the sides passing through the origin makes an angle $30°$ with the positive direction of the x-axis, then the sum of the x-coordinates of the vertices of the square is :

(A) $\sqrt{3}-2$ (B) $\sqrt{3}-1$ (C) $2\sqrt{3}-1$ (D) $2\sqrt{3}-2$

MSL233

For More Join: @IITJEE_Advanced

8. The point (2, 1) is translated parallel to the line $L : x - y = 4$ by $2\sqrt{3}$ units. If the new point Q lies in the third quadrant, then the equation of the line passing through Q and perpendicular to L is:

(A) $x + y = 3 - 3\sqrt{6}$ (B) $x + y = 3 - 2\sqrt{6}$

(C) $x + y = 2 - \sqrt{6}$ (4) $2x + 2y = 1 - \sqrt{6}$

MSL234

Subjective Type Questions:

9. Reduce the line $2x - 3y + 5 = 0$,

(a) In slope- intercept form and hence find slope & Y-intercept

MSL235

(b) In intercept form and hence find intercepts on the axes.

MSL236

(c) In normal form and hence find perpendicular distance from the origin and angle made by the perpendicular with the positive x-axis.

MSL237

10. Find distance of point $A(2, 3)$ measured parallel to the line $x - y = 5$ from the line $2x + y + 6 = 0$.

MSL238

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	D	A	A	D	D	D	D	B

9. (a) $y = \frac{2x}{3} + \frac{5}{3}, \frac{2}{3}, \frac{5}{3}$; (b) $\frac{x}{(-5/2)} + \frac{y}{(5/3)} = 1, -\frac{5}{2}, \frac{5}{3}$;

(c) $-\frac{2x}{\sqrt{13}} + \frac{3y}{\sqrt{13}} = \frac{5}{\sqrt{13}}, \frac{5}{\sqrt{13}}, \alpha = \pi - \tan^{-1}\left(\frac{3}{2}\right)$

10. $13\sqrt{2}/3$ units

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Straight Line: RACE-07

Single Correct Type Questions:

1. Consider the straight lines
$L_1: x - y = 1$
$L_2: x + y = 1$
$L_3: 2x + 2y = 5$
$L_4: 2x - 2y = 7$
The correct statement is :
(A) $L_1 \perp L_2, L_2 \parallel L_3, L_1$ intersects L_4
(B) $L_1 \perp L_2, L_1 \parallel L_3, L_1$ intersects L_2
(C) $L_1 \perp L_2, L_1 \perp L_3, L_2$ intersects L_4
(D) $L_1 \parallel L_4, L_2 \parallel L_3, L_2$ intersects L_3

MSL239

2. The point of intersection of the lines $(a^3 + 3)x + ay + a - 3 = 0$ and $(a^5 + 2)x + (a + 2)y + 2a + 3 = 0$ (a real) lies on the y-axis for :-
(A) No value of a
(B) Exactly two values of a
(C) More than two values of a
(D) Exactly one value of a

MSL240

3. The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2 x + (p^2 + 1)y + 2q = 0$ are perpendicular to a common line for :
(A) exactly one values of p
(B) exactly two values of p
(C) more than two values of p
(D) no value of p

Bankers Academy JEE

MSL241

Subjective Type Questions:

4. Find the angle between the lines $3x + y - 7 = 0$ and $x + 2y - 9 = 0$. **MSL242**

5. Find the line passing through the point (2, 3) and perpendicular to the straight line $4x - 3y = 10$.

MSL243

6. Find the equation of the line which has positive y-intercept 4 units and is parallel to the line $2x - 3y - 7 = 0$. Also find the point where it cuts the x-axis.

MSL244

7. Classify the following pairs of lines as coincident, parallel or intersecting:
(a) $x + 2y - 3 = 0$ & $-3x - 6y + 9 = 0$
(b) $x + 2y + 1 = 0$ & $2x + 4y + 3 = 0$
(c) $3x - 2y + 5 = 0$ & $2x + y - 5 = 0$

MSL245

8. A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L & the coordinate axes is 5. Find the equation of the line.

MSL246

ANSWER KEY

1. C **2.** A **3.** A **4.** $\theta = 135°$ or $45°$ **5.** $3x + 4y = 18$
6. $2x - 3y + 12 = 0, (-6, 0)$ **7.** (a) Coincident, (b) Parallel, (c) Intersecting
8. $x + 5y + 5\sqrt{2} = 0$ or $x + 5y - 5\sqrt{2} = 0$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Straight Line: RACE-08

Single Correct Type Questions:

1. The set of all possible values of θ in the interval $(0, \pi)$ for which the points $(1, 2)$ and $(\sin\theta, \cos\theta)$ lie on the same side of the line $x + y = 1$ is :

(A) $\left(0, \frac{\pi}{4}\right)$ (B) $\left(0, \frac{3\pi}{4}\right)$ (C) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (D) $\left(0, \frac{\pi}{2}\right)$

MSL247

2. If the point $(1, a)$ lies in between the straight lines $x + y = 1$ and $2(x + y) = 3$ then a lies in interval :-

(A) $\left(1, \frac{3}{2}\right)$ (B) $\left(0, \frac{1}{2}\right)$ (C) $(-\infty, 0)$ (D) $\left(\frac{3}{2}, \infty\right)$

MSL248

3. If (a, a^2) falls inside the angle made by the lines $y = \frac{x}{2}, x > 0$ and $y = 3x, \ x > 0$, then a belong to

(A) $\left(0, \frac{1}{2}\right)$ (B) $(3, \infty)$ (C) $\left(\frac{1}{2}, 3\right)$ (D) $\left(-3, -\frac{1}{2}\right)$

MSL249

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4. A straight line through origin O meets the lines $3y = 10 - 4x$ and $8x + 6y + 5 = 0$ at points A and B respectively. Then O divides the segment AB in the ratio

(A) 2 : 3 (B) 1 : 2 (C) 4 : 1 (D) 3 : 4

MSL250

5. The base of an equilateral triangle is along the line given by $3x + 4y = 9$. If a vertex of the triangle is $(1, 2)$, then the length of a side of the triangle is :

(A) $\frac{4\sqrt{3}}{15}$ (B) $\frac{4\sqrt{3}}{5}$ (C) $\frac{2\sqrt{3}}{15}$ (D) $\frac{2\sqrt{3}}{5}$

MSL251

Integer Type Questions:

6. If the line, $2x - y + 3 = 0$ is at a distance $\frac{1}{\sqrt{5}}$ and $\frac{2}{\sqrt{5}}$ from the lines $4x - 2y + \alpha = 0$ and $6x - 3y + \beta = 0$, respectively, then the sum of all possible values of α and β is ____

MSL252

Subjective Type Questions:

7. Examine the positions of the points $(3, 4)$ and $(2, -6)$ w.r.t. $3x - 4y = 8$

MSL253

8. If $(2, 9)$, $(-2, 1)$ and $(1, -3)$ are the vertices of a triangle, then prove that the origin lies inside the triangle.

MSL254

For More Join: @IITJEE_Advanced

9. Find the distances between the following pair of parallel lines :
(a) $3x + 4y = 13,\ 3x + 4y = 3$
(b) $3x - 4y + 9 = 0,\ 6x - 8y - 15 = 0$

MSL255

10. Find the points on the x-axis such that their perpendicular distance from the line $\frac{x}{a}+\frac{y}{b} = 1$ is 'a', $a, b > 0$.

MSL256

11. Show that the area of the parallelogram formed by the lines
$2x - 3y + a = 0, 3x - 2y - a = 0, 2x - 3y + 3a = 0$ and $3x - 2y - 2a = 0$ is $\frac{2a^2}{5}$ square units.

MSL257

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6
Answer	D	B	C	C	A	30

7. opposite sides of the line **9.** (a) 2; (b) $\frac{33}{10}$ **10.** $\left(\frac{a}{b}\left[b \pm \sqrt{a^2+b^2}\right], 0\right)$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Straight Line: RACE-09

Single Correct Type Questions:

1. Let L denote the line in the xy-plane with x and y intercepts as 3 and 1 respectively. Then the image of the point (–1, –4) in this line is :

(A) $\left(\frac{8}{5}, \frac{29}{5}\right)$ (B) $\left(\frac{29}{5}, \frac{11}{5}\right)$ (C) $\left(\frac{11}{5}, \frac{28}{5}\right)$ (D) $\left(\frac{29}{5}, \frac{8}{5}\right)$

MSL258

2. A ray of light is incident along a line which meets another line, $7x - y + 1 = 0$, at the point (0, 1). The ray is then reflected from this point along the line, $y + 2x = 1$. Then the equation of the line of incidence of the ray of light is :-

(A) $41x - 25y + 25 = 0$ (B) $41x + 25y - 25 = 0$
(C) $41x + 38y - 38 = 0$ (D) $41x - 38y + 38 = 0$

MSL259

3. Let L be the line passing through the point $P(1, 2)$ such that its intercepted segment between the co-ordinate axes is bisected at P. If L_1 is the line perpendicular to L and passing through the point (–2, 1), then the point of intersection of L and L_1 is :

(A) $\left(\frac{3}{10}, \frac{17}{5}\right)$ (B) $\left(\frac{11}{20}, \frac{29}{10}\right)$ (C) $\left(\frac{4}{5}, \frac{12}{5}\right)$ (D) $\left(\frac{3}{5}, \frac{23}{10}\right)$

Pankaj Academy JEE

MSL260

4. The foot of the perpendicular drawn from the origin, on the line, $3x + y = \lambda(\lambda \neq 0)$ is P. If the line meets x-axis at A and y–axis at B, then the ratio $BP : PA$ is :-

(A) 9 : 1 (B) 1 : 3 (C) 3 : 1 (D) 1 : 9

MSL261

5. A ray of light passing through the point $A(1,2)$ is reflected at a point B on the x-axis and then passes through (5,3). Then the equation of AB is :

(A) $5x + 4y = 13$ (B) $5x - 4y = -3$
(C) $4x + 5y = 14$ (D) $4x - 5y = -6$

MSL262

6. A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x-axis, the equation of the reflected ray is :

(A) $y = x + \sqrt{3}$ (B) $\sqrt{3}y = x - \sqrt{3}$
(C) $y = \sqrt{3}x - \sqrt{3}$ (D) $\sqrt{3}y = x - 1$

MSL263

7. If the image of point $P(2, 3)$ in a line L is $Q(4, 5)$ then, the image of point $R(0, 0)$ in the same line is :

(A) (4, 5) (B) (2, 2) (C) (3, 4) (D) (7, 7)

MSL264

For More Join: @IITJEE_Advanced

Integer Type Questions:

8. Let P be the point (3, 2). Let Q be the reflection of P about the x-axis. Let R be the reflection of Q about the line $y = -x$ and let S be the reflection of R through the origin. $PQRS$ is a convex quadrilateral. Find the area of $PQRS$.

MSL265

Matrix Type Questions:

9. The point (4, 1) undergoes the following transformations, then the match the correct alternatives:

	Column-I		**Column-II**
(A)	Reflection about x-axis is	(p)	(4, –1)
(B)	Reflection about y-axis is	(q)	(–4, –1)
(C)	Reflection about origin is	(r)	$\left(-\frac{12}{25}, -\frac{59}{25}\right)$
(D)	Reflection about the line $y = x$ is	(s)	(–4, 1)
(E)	Reflection about the line $4x + 3y - 5 = 0$ is	(t)	(1, 4)

MSL266

Subjective Type Questions:

10. The equations of the perpendicular bisectors of the sides AB & AC of a triangle ABC are $x - y + 5 = 0$ & $x + 2y = 0$, respectively. If the point A is (1, –2) find the equation of the line BC.

MSL267

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	C	D	C	A	A	B	D	15	A → p; B → s; C → q; D → t; E → r

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Straight Line: RACE-10

Single Correct Type Questions:

1. If the three distinct lines $x + 2ay + a = 0, x + 3by + b = 0$ and $x + 4ay + a = 0$ are concurrent, then the point (a, b) lies on a :-

(A) circle (B) straight line (C) parabola (D) hyperbola

MSL268

2. The line parallel to x-axis and passing through the point of intersection of the lines $ax + 2by + 3b = 0$ and $bx - 2ay - 3a = 0$, where $(a, b) \neq (0, 0)$ is :

(A) below x-axis at a distance $2/3$ from it (B) above x-axis at a distance $3/2$ from it

(C) above x-axis at a distance $2/3$ from it (D) below x-axis at a distance $3/2$ from it

MSL269

3. Consider the set of all lines $px + qy + r = 0$ such that $3p + 2q + 4r = 0$. Which one of the following statements is true ?

(A) The lines are all parallel. (B) Each line passes through the origin.

(C) The lines are not concurrent (D) The lines are concurrent at the point $\left(\frac{3}{4}, \frac{1}{2}\right)$

MSL270

Subjective Type Questions:

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4. To what point must the origin be shifted, so that the coordinates of the point (4, 5) become (– 3, 9).

MSL271

5. If the axes be turned through an angle $\tan^{-1}2$ (in anticlockwise direction), what does the equation $4xy - 3x^2 = a^2$ become?

MSL272

6. Find the value of λ, if the lines $3x - 4y - 13 = 0, 8x - 11y - 33 = 0$ and $2x - 3y + \lambda = 0$ are concurrent.

MSL273

7. Find the equation of the line joining the point (2, – 9) and the point of intersection of lines $2x + 5y - 8 = 0$ and $3x - 4y - 35 = 0$.

MSL274

8. Find the equations of the line which pass through the point of intersection of the lines $4x - 3y = 1$ and $2x - 5y + 3 = 0$ and is equally inclined to the coordinate axes.

MSL275

9. Find the equation of the line through the point of intersection of the lines $3x - 4y + 1 = 0$ & $5x + y - 1 = 0$ and perpendicular to the line $2x - 3y = 10$.

MSL276

ANSWER KEY

1.	B	**2.**	D	**3.**	D	**4.**	$7, -4$	**5.**	$x^2 - 4y^2 = a^2$
6.	$\lambda = -7$	**7.**	$-y + x = 11$	**8.**	$x + y = 2, x = y$			**9.**	$69x + 46y - 25 = 0$

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Straight Line: RACE-11

Single Correct Type Questions:

1. Let $P = (-1, 0)$, $Q = (0, 0)$ and $R = \left(3, 3\sqrt{3}\right)$ be three points. The equation of the bisector of the angle PQR is

(A) $\frac{\sqrt{3}}{2}x + y = 0$ (B) $x + \sqrt{3y} = 0$

(C) $\sqrt{3}x + y = 0$ (D) $x + \frac{\sqrt{3}}{2}y = 0$

MSL277

2. If a variable line drawn through the intersection of the lines $\frac{x}{3} + \frac{y}{4} = 1$ and $\frac{x}{4} + \frac{y}{3} = 1$, meets the coordinate axes at A and B, $(A \neq B)$, then the locus of the midpoint of AB is :-

(A) $14(x + y)^2 - 97(x + y) + 168 = 0$

(B) $6xy = 7(x + y)$

(C) $7xy = 6(x + y)$

(D) $4(x + y)^2 - 28(x + y) + 49 = 0$

Rankers Academy JEE

MSL278

3. The sides of a rhombus $ABCD$ are parallel to the lines, $x - y + 2 = 0$ and $7x - y + 3 = 0$. If the diagonals of the rhombus intersect at $P(1,2)$ and the vertex A (different from the origin) is on the y–axis, then the ordinate of A is :-

(A) 2 (B) $\frac{5}{2}$

(C) $\frac{7}{4}$ (D) $\frac{7}{2}$

MSL279

4. A square of side a lies above the x-axis and has one vertex at the origin. The side passing through the origin makes an angle $\alpha\left(0 < \alpha < \frac{\pi}{4}\right)$ with the positive direction of x-axis. The equation of its diagonal not passing through the origin is

(A) $y(\cos\alpha + \sin\alpha) + x(\cos\alpha - \sin\alpha) = a$

(B) $y(\cos\alpha - \sin\alpha) - x(\sin\alpha - \cos\alpha) = a$

(C) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha - \cos\alpha) = a$

(D) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha + \cos\alpha) = a$

MSL280

For More Join: @IITJEE_Advanced

Subjective Type Questions:

5. Find the equations of bisectors of the angle between the lines $4x + 3y = 7$ and $24x + 7y - 31 = 0$. Also find which of them is
(a) the bisector of the angle containing origin
(b) the bisector of the acute angle.

MSL281

6. Given vertices $A(1, 1)$, $B(4, -2)$ & $C(5, 5)$ of a triangle, find the equation of the perpendicular dropped from C to the interior bisector of the angle A.

MSL282

7. Consider the family of lines $(x - y - 6) + \lambda(2x + y + 3) = 0$ and $(x + 2y - 4) + \lambda'(3x - 2y - 4) = 0$. If the lines of these 2 families are at right angle to each other then find the locus of their point of intersection.

MSL283

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4
Answer	C	C	B	A

5. (a) $x - 2y + 1 = 0$; (b) $2x + y - 3 = 0$ **6.** $x - 5 = 0$

7. $x^2 + y^2 - 3x + 4y - 3 = 0$

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Straight Line: RACE-12

Single Correct Type Questions:

1. If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is

(A) 1 (B) 2 (C) –1/2 (D) –2

MSL284

2. If one of the lines given by $6x^2 - xy + 4cy^2 = 0$ is $3x + 4y = 0$, then c equals

(A) –3 (B) 1 (C) 3 (D) 1

MSL285

3. If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then c has the value

(A) –2 (B) –1 (C) 2 (D) 1

MSL286

4. If the pair of straight lines $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ be such that each pair bisects the angle between the other pair, then

(A) $pq = -1$ (B) $p = q$ (C) $p = -q$ (D) $pq = 1$

MSL287

5. The pair of lines represented by $3ax^2 + 5xy + (a^2 - 2)y^2 = 0$ are perpendicular to each other for

(A) two values of a (B) $\forall\ a$

(C) for one value of a (D) for no values of a

MSL288

6. If the pair of lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ intersect on the y-axis then

(A) $2fgh = bg^2 + ch^2$ (B) $bg^2 \neq ch^2$

(C) $abc = 2fgh$ (D) none of these

MSL289

7. If the line $y = mx$ bisects the angle between the lines $ax^2 + 2h\,xy + by^2 = 0$ then m is a root of the quadratic equation:

(A) $hx^2 + (a - b)x - h = 0$ (B) $x^2 + h(a - b)x - 1 = 0$

(C) $(a - b)x^2 + hx - (a - b) = 0$ (D) $(a - b)x^2 - hx - (a - b) = 0$

MSL290

8. The greatest slope along the graph represented by the equation $4x^2 - y^2 + 2y - 1 = 0$, is-

(A) –3 (B) –2 (C) 2 (D) 3

MSL291

Subjective Type Questions:

9. Prove that the equation $x^2 - 5xy + 4y^2 = 0$ represents two lines passing through the origin. Also find their equations.

MSL292

10. If the equation $3x^2 + kxy - 10y^2 + 7x + 19y = 6$ represents a pair of lines, find the value of k.

MSL293

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For More Join: @IITJEE_Advanced

11. If the equation $6x^2 - 11xy - 10y^2 - 19y - 6 = 0$ represents a pair of lines, find their equations. Also find the angle between the two lines.

MSL294

12. Find the point of intersection and the angle between the lines given by the equation :
$2x^2 - 3xy - 2y^2 + 10x + 5y = 0$.

MSL295

13. Consider a line pair $2x^2 + 3xy - 2y^2 - 10x + 15y - 28 = 0$ and another line L passing through origin with gradient 3. The line pair and line L form a triangle whose vertices are A, B and C.
(a) Find the sum of the cotangents of the interior angles of the triangle ABC.

MSL296

(b) Find the area of triangle ABC

MSL297

(c) Find the radius of the circle touching all the 3 sides internally of the triangle.

MSL298

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	A	A	C	A	A	A	A	C

9. $x - y = 0$ & $x - 4y = 0$

10. $k = -1$, or $-\frac{127}{6}$

11. $2x - 5y - 2 = 0$ & $3x + 2y + 3 = 0$; $\tan^{-1}\left(\frac{19}{4}\right)$

12. $(-1, 2)$, $90°$

13. (a) $\frac{50}{7}$; (b) $\frac{63}{10}$; (c) $\frac{3}{10}\left(8\sqrt{5} - 5\sqrt{10}\right)$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Straight Line: RACE-13

Single Correct Type Questions:

1. Find the angle subtended at the origin by the intercept made on the curve $x^2 - y^2 - xy + 3x - 6y + 18 = 0$ by the line $2x - y = 3$.

MSL299

2. Find the equation of the lines joining the origin to the points of intersection of the curve $2x^2 + 3xy - 4x + 1 = 0$ and the line $3x + y = 1$.

MSL300

3. A straight line is drawn from the point (1, 0) to the curve $x^2 + y^2 + 6x - 10y + 1 = 0$, such that the intercept made on it by the curve subtends a right angle at the origin. Find the equations of the line.

MSL301

4. Find the equation to the pair of straight lines joining the origin to the intersections of the straight line $y = mx + c$ and the curve $x^2 + y^2 = a^2$. Prove that they are at right angles if $2c^2 = a^2(1 + m^2)$.

MSL302

5. Prove that the straight lines joining the origin to the points of intersection of the straight line $kx + hy = 2hk$ with the curve $(x - h)^2 + (y - k)^2 = c^2$ are at right angles if $h^2 + k^2 = c^2$.

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MSL303

6. If line $2x - y - 1 = 0$ intersect the curve $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ in two points A and B, such that $\angle AOB = 90°$, then $\left|\frac{a+b+5c}{2g-f}\right|$ is equal to

MSL304

ANSWER KEY

1. $\theta = \tan^{-1}\frac{4}{7}$ **2.** $x^2 - y^2 - 5xy = 0$ **3.** $x + y = 1;\ x + 9y = 1$

4. $x^2 + y^2 - a^2\left(\frac{y - mx}{c}\right)^2 = 0$ **6.** 2

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